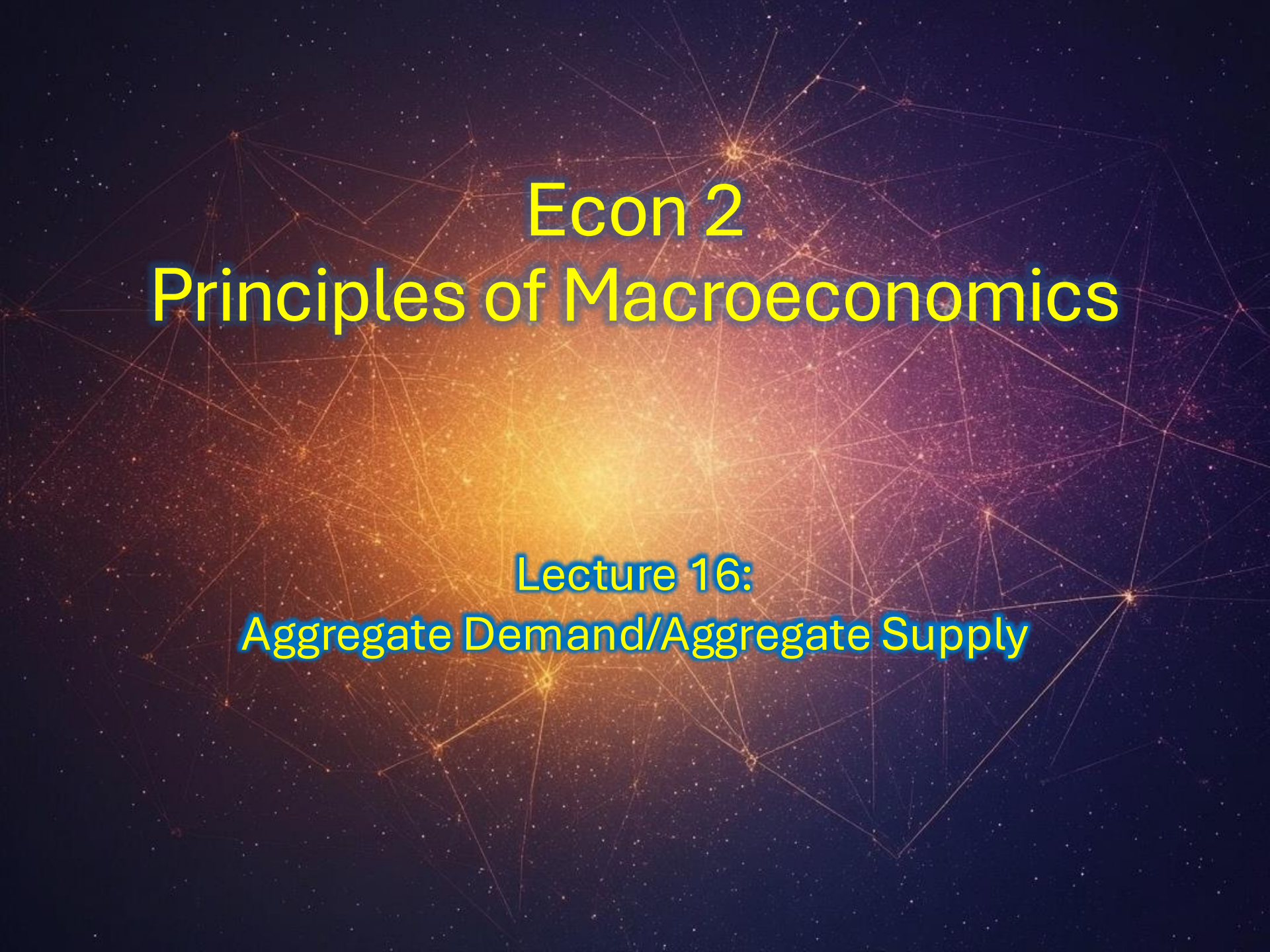




Econ 2

Principles of Macroeconomics

Lecture 16:

Aggregate Demand/Aggregate Supply

Course Evaluations

- Course is being updated
- Textbook
- AI Integration
- Gamification Ideas



PRIN ECON-MACRO

Student-FO

https://go.blueja.io/VRz0y1-1-0G_Fvd6fWEn8w

Why is the interest rate so important?

- What changes when r decreases?

- 1.

- 2.

- 3.



Monetary Policy in Action

- Example 1: $Y_1 = 8,000$, $r_1 = 6\%$ and $M_1 = 1$ trillion
- $\bar{Y} = 10,000$ (full employment GDP)



Monetary Policy in Action

- Example 1: $Y_3 = 12,000$, $r_3 = 2\%$ and $M_3 = 4$ trillion
- $\bar{Y} = 10,000$ (full employment GDP)



Effects of Inflation

- Example: $\bar{Y} = 10,000$ (full employment GDP), $r_1 = 5\%$, $M_1 = 1$ trillion, $CPI_1 = 100$



Aggregate Demand Curve

- How are prices and levels of $Y=AE$ related?



- Aggregate Demand Curve: “Equilibrium Level of Spending at Every Price Level” Curve
As general level of prices rise, equilibrium level of spending ($Y^*=AE^*$) decreases
- Moving along the aggregate demand curve?

Aggregate Demand Curve

- Shifts in the AD Curve



Aggregate Supply Curve

- Aggregate Supply: The amount that firms are willing and able to produce at various price levels
- In the short-run:
 - firms respond to the costs of production
 - production costs are determined by level of Y in short-run
 - as Y increases, new (less productive) workers are hired
 - additional Y costs more than previous Y



Aggregate Supply Curve

- Movement Along vs. Shifts in Supply
- What shifts AS?
- Anything that changes the amount firms are willing to produce that is not the price
- Examples:

Complete Macroeconomic Equilibrium



Entire Macroeconomy

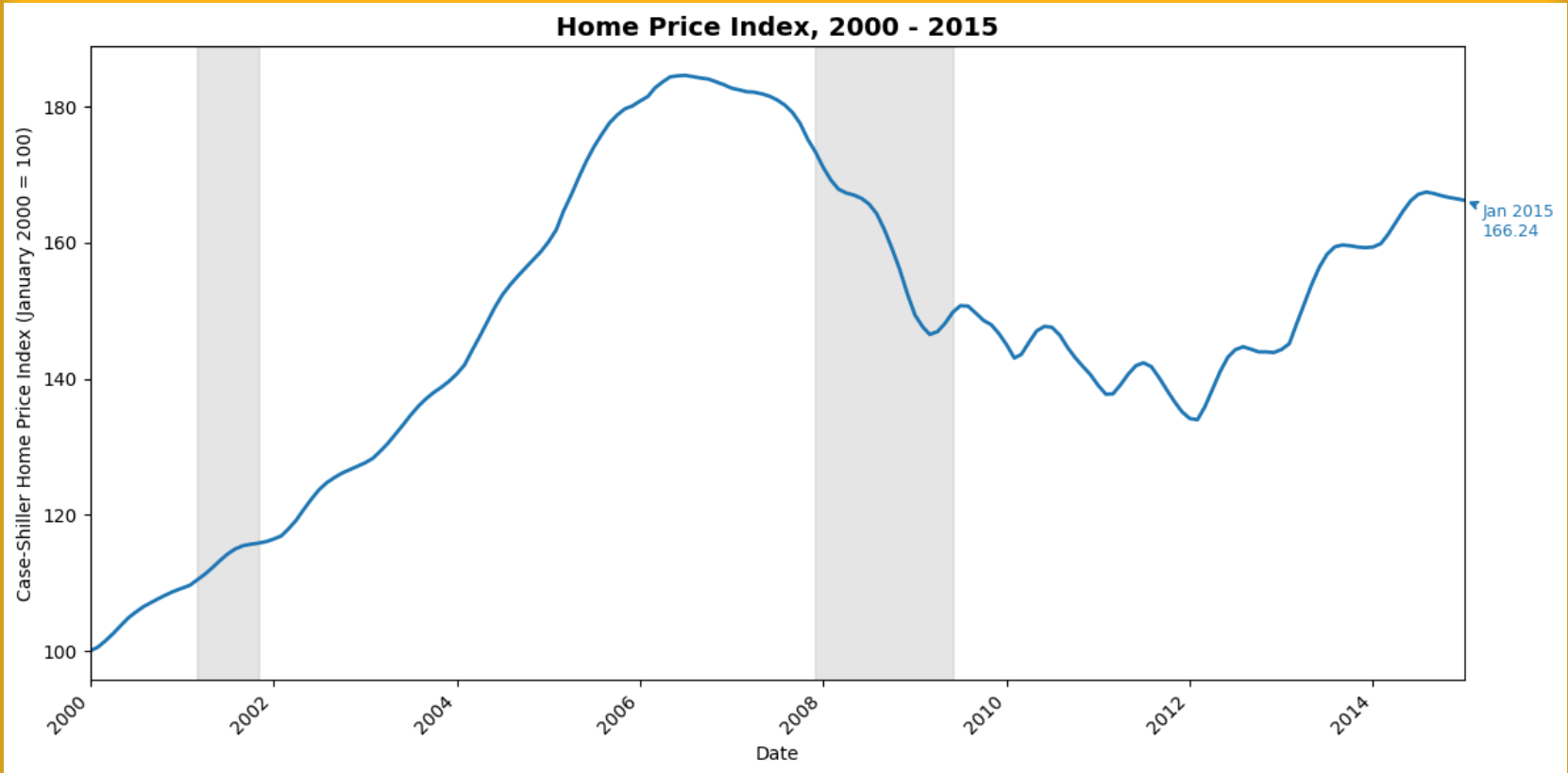


- Money Supply = 1 trillion
- Interest Rate = 5%
- Price Level = 100
- $Y = \bar{Y} = 10,000$

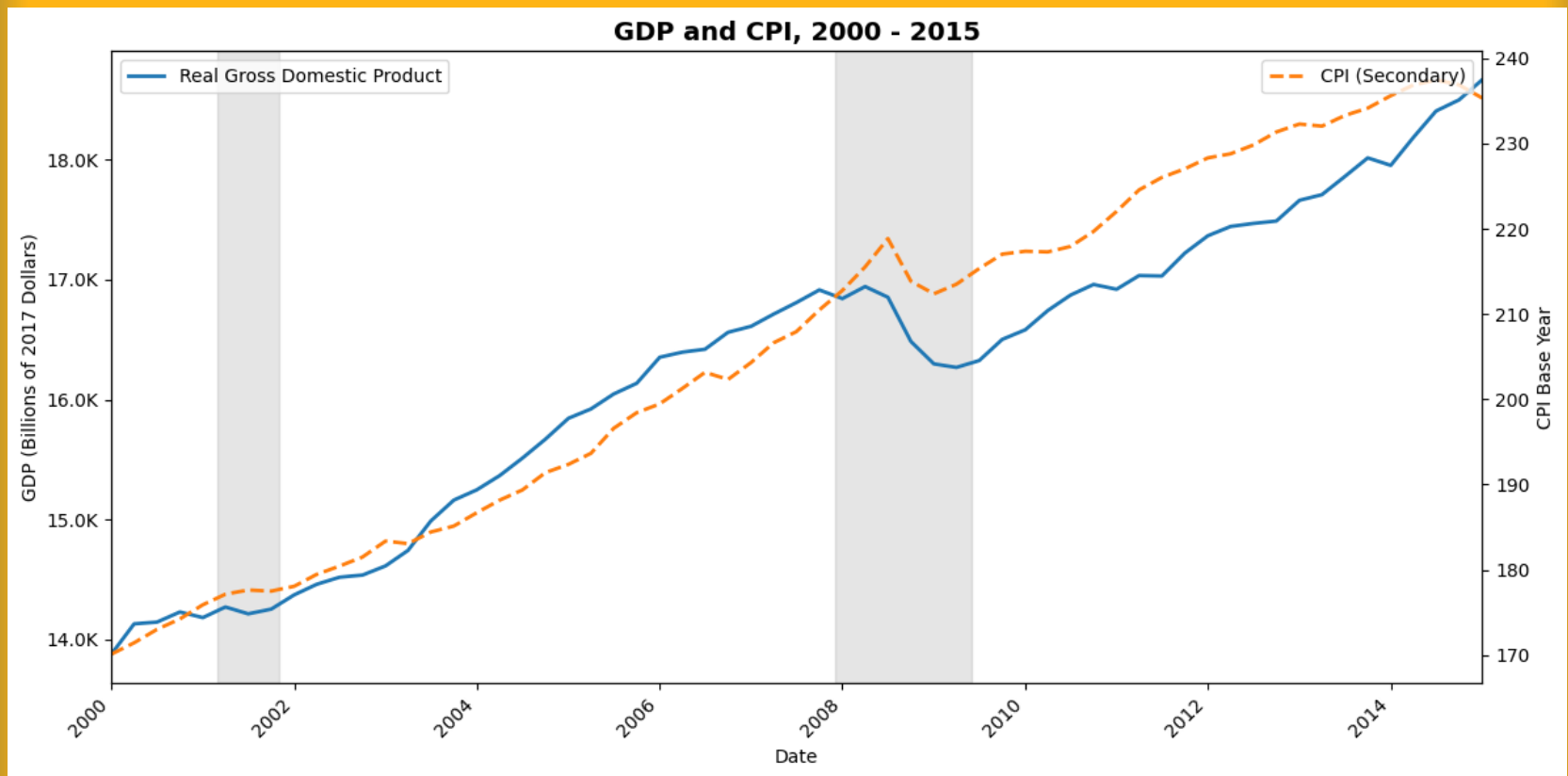
Shocks to the Macroeconomy

- Connect model to real-world data
- During the Great Recession, is there evidence that:
 - housing prices fell
 - output and CPI decreased
 - government spending increased
 - tax revenue decreased
 - interest rates decreased (money supply increased)

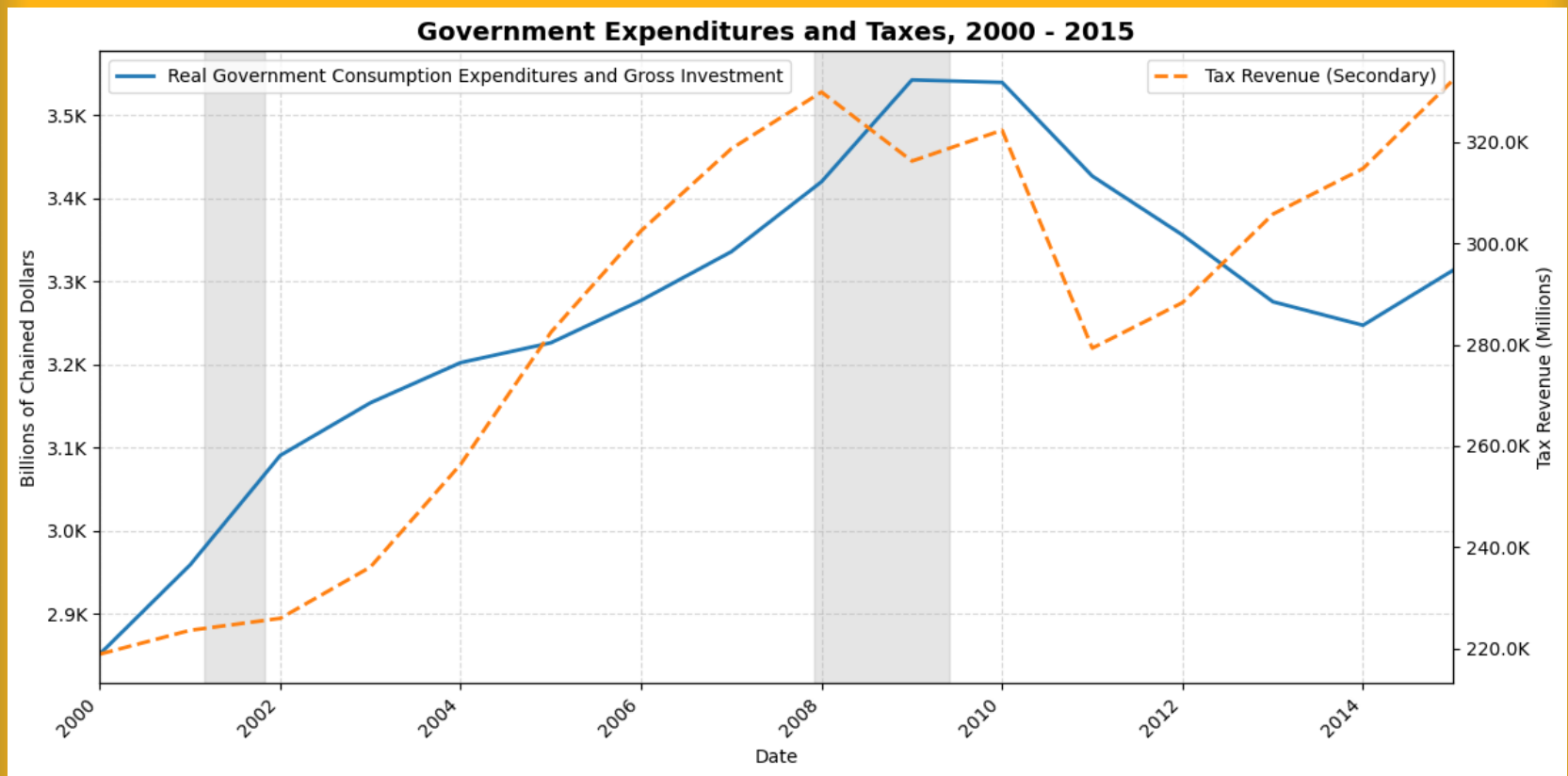
Housing Prices



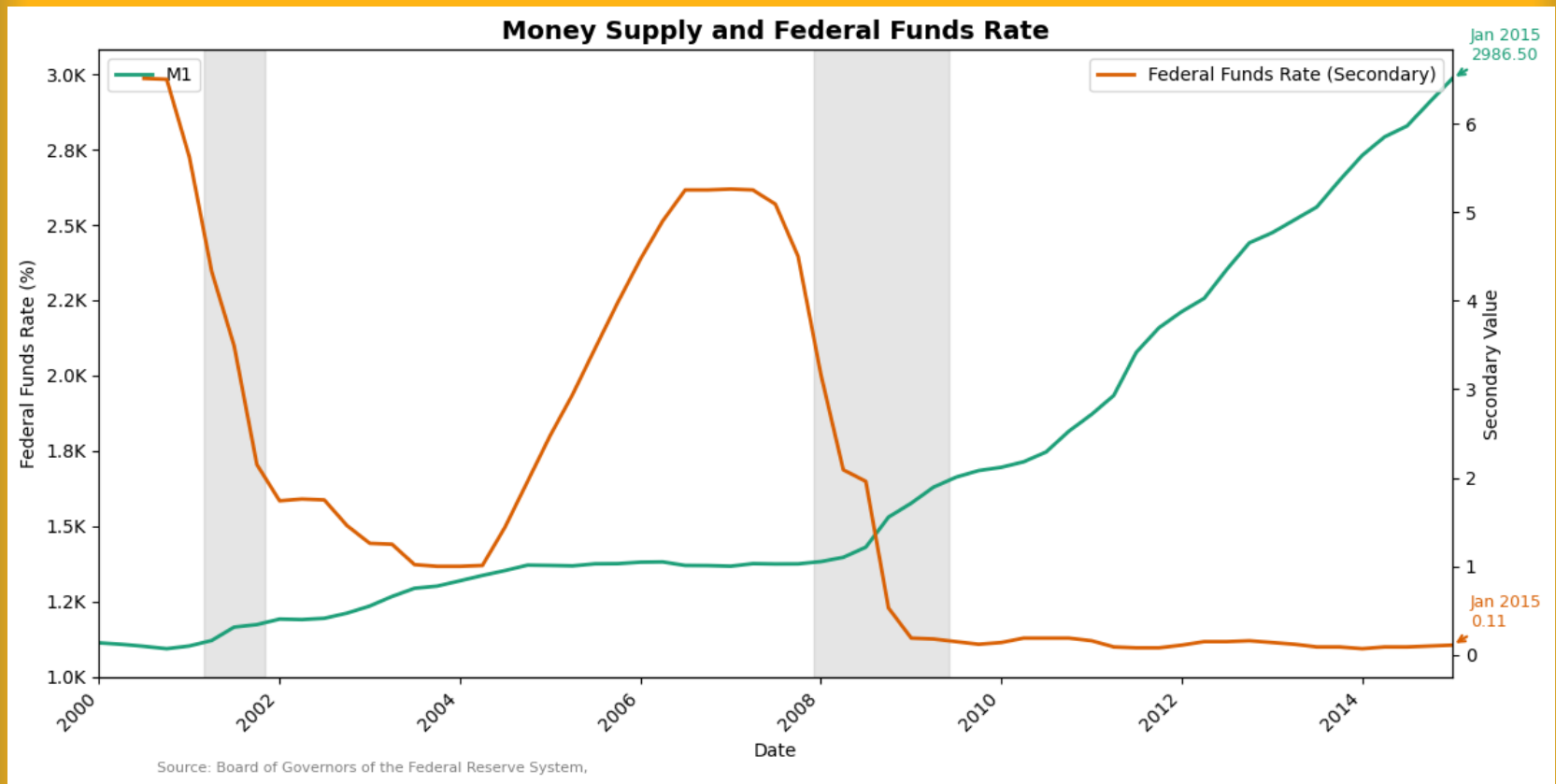
GDP/CPI



Taxes and Government Expenditures



Interest Rate and Money Supply



Shocks to the Macroeconomy

- Start at Full-Employment output
- Housing Market Crash-how does P^* and Y^* change?



Shocks to the Macroeconomy

- Start at Full-Employment output
- Housing Market Crash-how does P^* and Y^* change?
- How should the government or Fed respond?



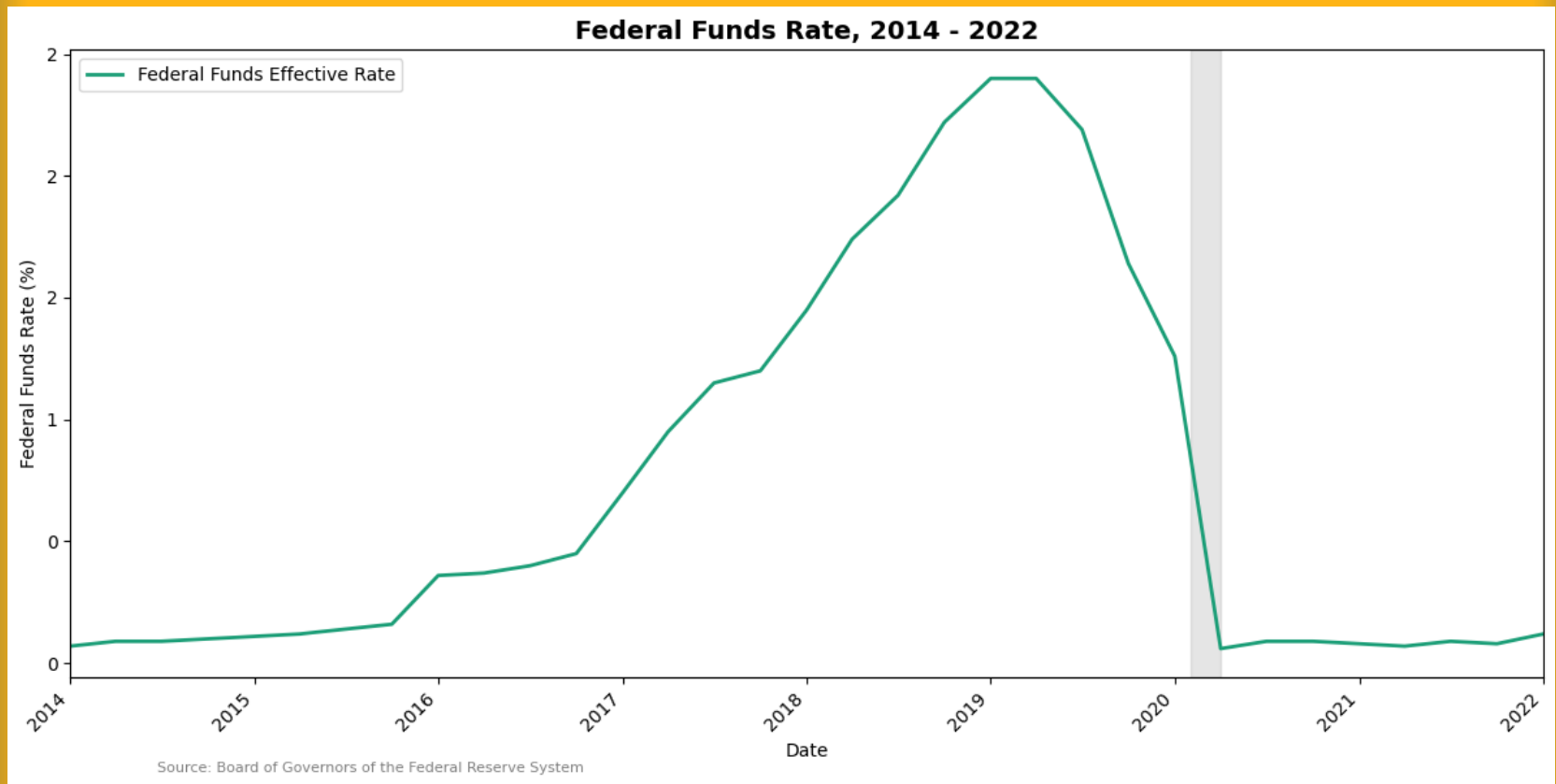
Demand Shocks to the Macroeconomy

- Demand Shock:
- Counter-Cyclical Policy:

The Federal Reserve During COVID

- Emergency Rate Cut I on March 3rd, 2020
 - Fed Funds Rate: 1.0 to 1.25%
- Emergency Rate Cut II on March 15th, 2020
 - Fed Funds Rate: 0 to 0.25%
- Only unconventional tools remain
 - Buying mortgage-backed securities
 - Lower reserve requirements for banks
- All meant to increase AC, I^P

Federal Funds Rate, 2014-2022



Supply Shocks

- Supply Shock: Change in AS that moves economy away from full-employment equilibrium
- How do we know if a recession is caused by a supply shock instead of demand shock?



Supply Shock

- How should the government respond to a recession driven by a decrease in AS?
- Option 1: Fight unemployment, bring economy back to $Y\text{-bar}$

Supply Shock

- How should the government respond to a recession driven by a decrease in AS?
- Option 2: Fight rising prices, bring prices back to P^*

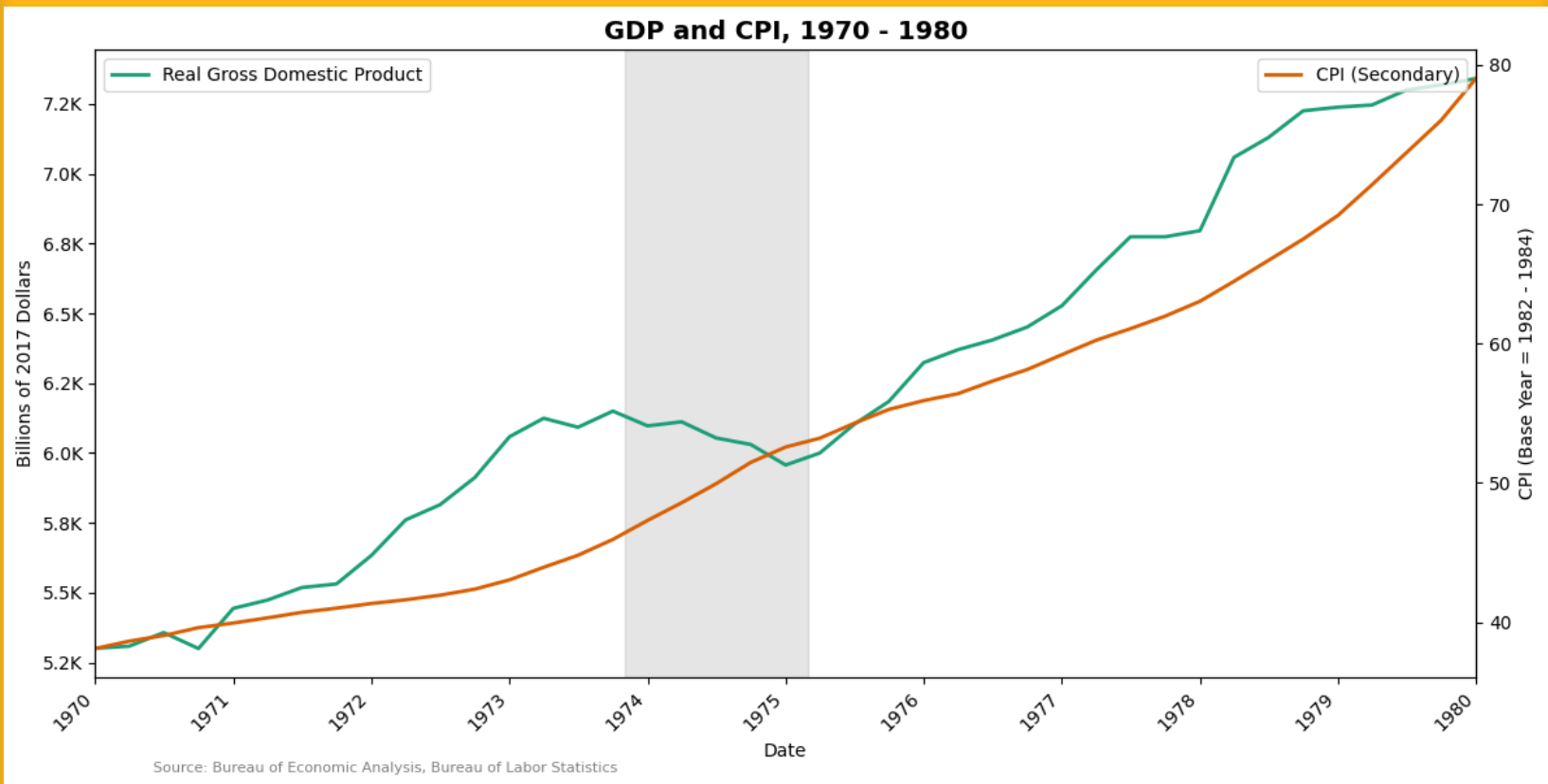
Historical Supply Shocks

- OPEC Oil Embargo in the early 1970s
- Microeconomic Consequences: Gasoline shortage



Historical Supply Shocks

- Did prices rise as output fell?



Historical Supply Shocks

- How did the Federal Reserve respond?

